9. Array Minimise the maximum difference between heights [V.IMP]

```cpp
class Solution {
public:
  int getMinDiff(vector<int> &arr, int k) {
    int n = arr.size();
    sort(arr.begin(), arr.end());

    int diff = arr[n-1] - arr[0]; // initial difference

    int smallest = arr[0] + k;
    int largest = arr[n-1] - k;

    for(int i = 1; i < n; i++) {
      int min_height = min(smallest, arr[i] - k);
      int max_height = max(largest, arr[i-1] + k);

      if(min_height < 0) continue; // skip if negative

      diff = min(diff, max_height - min_height);
    }

    return diff;
  }
};
```